

House Price Prediction

A regression-based analysis of residential property values — identifying which structural, amenity, and locational features drive price, and by how much.

REGRESSION ANALYSIS

545 PROPERTIES

13 VARIABLES



Made with GAMMA



Why Predicting House Prices Matters

Reduce Information Asymmetry

Sellers avoid under/overpricing; buyers validate fair asking prices; platforms automate valuations at scale.

Statistically Grounded Benchmark

Quantified evidence of which features most influence price, supporting renovation and investment prioritization.

Transparent & Auditable

An interpretable model — not a black box — that underwriters and appraisers can review and trust.

Dataset at a Glance



Source: Housing.csv

545 observations · 13 columns (1 target + 12 features) · Zero missing values · Zero duplicates

Numerical Features

area, bedrooms, bathrooms, stories, parking

Binary Categorical

mainroad, guestroom, basement, hotwaterheating, airconditioning, prefarea

Multi-class

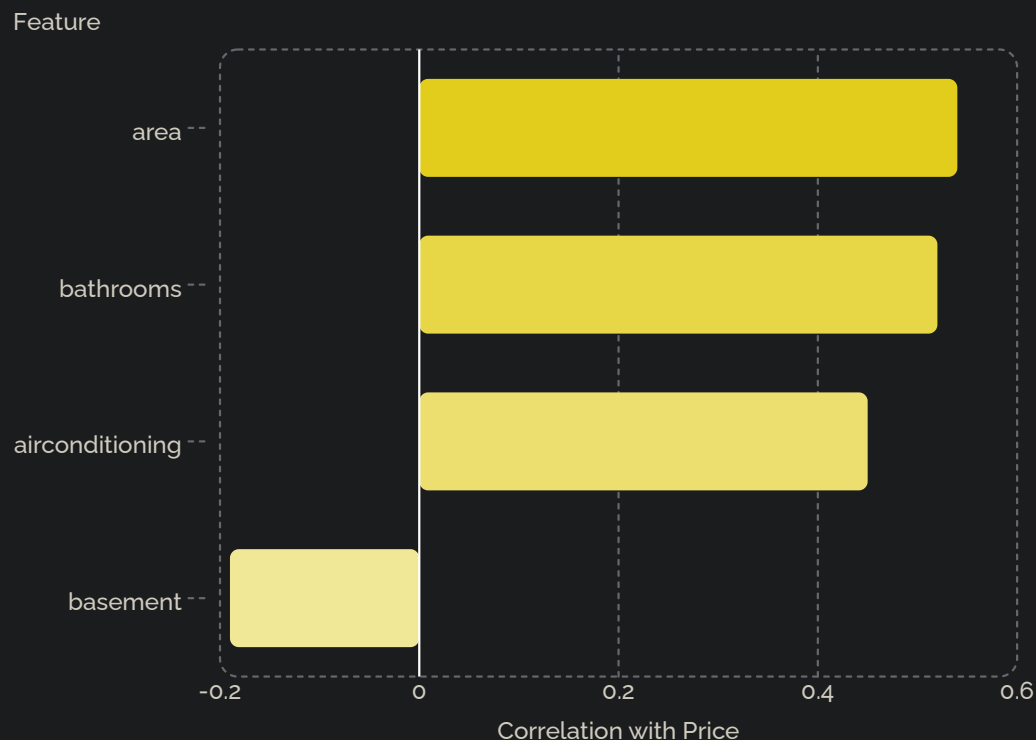
furnishingstatus: furnished / semi-furnished / unfurnished

Project Workflow

01	02	03
Business Understanding	Data Collection & Cleaning	Exploratory Data Analysis
04	05	06
Feature Engineering	Correlation & Multicollinearity	Regression Modeling
07	08	09
Model Evaluation	OLS & Refinement	Assumption Testing
10	11	
Log Transformation	Business Insights	

A structured pipeline progressing from raw data through diagnostics to a log-transformed model with superior statistical properties.

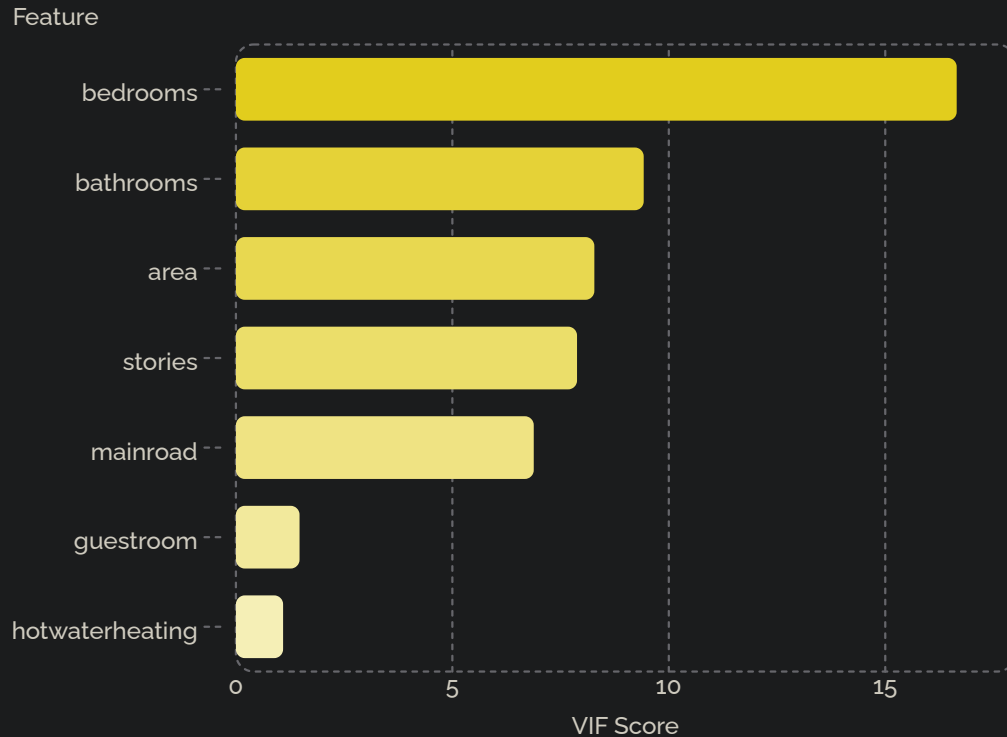
Exploratory Data Analysis: Key Findings



What the Data Revealed

- **Area, bathrooms, and AC** are the strongest positive correlates of price
- **Basement** shows a counterintuitive negative bivariate correlation (-0.19) — later reversed in the multivariate model
- Both **price** and **area** are right-skewed, motivating the later log-transformation
- Amenities like hot water heating and guest rooms are **rare**, making them market differentiators

Multicollinearity: VIF Analysis



Interpreting VIF

bedrooms (VIF = 16.65) shows severe multicollinearity — its effect is largely captured by area and bathrooms. **bathrooms**, **area**, **stories**, and **mainroad** show moderate VIF (5–10). All others are acceptable.

- High VIF flags potential redundancy but doesn't prove a variable is uninformative — OLS p-values confirmed bedrooms was insignificant before removal.

OLS Model Results: Full Feature Set

Overall model fit: $R^2 = 0.682$ · Adjusted $R^2 = 0.674$ · F-statistic p-value = $9.07e-123$

Feature	Coefficient	p-value	Business Meaning
bathrooms	+988K	0.000 ✓	Each additional bathroom adds ~988K to price
airconditioning	+865K	0.000 ✓	AC adds ~865K to price
hotwaterheating	+855K	0.000 ✓	Hot water heating adds ~855K
stories	+451K	0.000 ✓	Each additional story adds ~451K
prefarea	+652K	0.000 ✓	Preferred area adds ~652K
bedrooms	+115K	0.114 ☐	No reliable independent effect
furnishing_semi	-46K	0.691 ☐	No difference vs. furnished

Model Comparison: Three Versions

Criterion	Full Model	Refined Model	Log-Transformed ★
Features	13	11	11
R ² (test set)	0.6529	0.6529	0.6671
R ² (OLS)	0.682	—	0.696
Residual Normality	Violated (p=0.000)	Violated (p=0.000)	Not rejected (p=0.178)
Heteroscedasticity	Violated (p=3.49e-10)	Violated (p=3.49e-10)	Reduced (p=0.0073)
Durbin-Watson	1.209	1.209	1.292

- ✔ The **log-transformed model** is the preferred model: better residual normality, reduced heteroscedasticity, and improved R² — coefficients now interpretable as approximate percentage price changes.



Business Insights & Recommendations

Top Price Drivers

Bathrooms (+988K), air conditioning (+865K), and preferred area (+652K) are the most consistent, high-impact value levers across all model versions.

Counterintuitive Findings

Bedrooms show **no independent price effect** once area and bathrooms are controlled for. Basements add +350K in multivariate analysis despite a negative raw correlation.

For Investors & Developers

Prioritize adding bathrooms, AC, and parking over bedrooms. Treat basement-equipped properties as undervalued opportunities if the basement isn't reflected in comps.

Pricing Strategy

Use the log-transformed model as the primary reference — communicate price effects as percentage premiums (e.g., "+15% for AC") rather than flat currency estimates.

Limitations & Future Directions

Current Limitations

- No neighborhood-level location data beyond binary prefarea/mainroad flags
- No time dimension — cannot account for market trends or seasonality
- Modest dataset size (545 records) limits subgroup analysis
- Log model retains mild heteroscedasticity (BP $p=0.0073$) and autocorrelation (DW=1.292)
- ~30–35% of price variance unexplained — property condition, age, school districts, and macro factors absent

Future Improvements

- **Robust Regression** (Huber/RANSAC) to handle the 15 identified outliers
- **Tree-based ensembles** (Random Forest, XGBoost) to capture non-linear relationships
- **Interaction terms** (e.g., area × bathrooms) and polynomial features
- **K-fold cross-validation** for more stable performance estimates
- **Hyperparameter tuning** for Ridge/Lasso or gradient boosting models